

Reviewed Preprint

v1 • January 13, 2026

Not revised

Reviewed Preprint

v2 • March 11, 2026

Revised by authors

Prior cocaine use disrupts identification of hidden states by single units and neural ensembles in orbitofrontal cortex

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Competing interests: No

competing interests declared

Funding: See [page 20](#)

Reviewing editor: Michael A

McDannald, Boston College, United States

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eLife Assessment

This **fundamental** work shows that a history of cocaine self-administration disrupts the orbitofrontal cortex's ability to encode similarities between distinct sensory stimuli that possess identical task information - hidden states. The evidence supporting these conclusions is **compelling**, with methods and analyses spanning self-administration, a novel 'figure 8' sequential odor task, recordings from 3,881 single units, and sophisticated firing analyses revealing complex orbitofrontal representations of task structure. These results will be of broad interest to psychologists, neuroscientists, and clinicians.

<https://doi.org/10.7554/eLife.109883.2.sa3>

Abstract

The orbitofrontal cortex (OFC) is critical to identifying task structure and to generalizing appropriately across task states with similar underlying or hidden causes. This capability is at the heart of OFCs proposed role in a network responsible for cognitive mapping, and its loss can explain many deficits associated with OFC damage or inactivation. Substance use disorder is defined by behaviors that share much in common with these deficits, such as an inability to modify learned behaviors in the face of new information about undesired consequences. One explanation for this similarity would be if addictive drugs impacted the ability of OFC to recognize underlying similarities, hidden states, that allow information learned in one setting to be used in another. To explore this possibility, we trained rats to self-administer cocaine and then recorded single unit activity in lateral OFC as these rats performed in an odor sequence task consisting of unique and shared positions. In well-trained controls, we observed chance decoding of sequence at shared positions and near chance decoding even at unique positions, reflecting the irrelevance of distinguishing these positions in the task. By contrast, in cocaine-experienced rats, decoding remained significantly elevated, particularly at the positions that had superficial sensory differences that were collapsed in controls across learning. These neural differences were accompanied by increases in behavioral variability at these positions. A tensor component analysis showed that this effect of reduced generalization after cocaine use also extended across positions in the sequences. These results show that prior cocaine use disrupts the normal identification of hidden states by OFC.

Introduction

The orbitofrontal cortex (OFC) is essential for recognizing the underlying structure of tasks and for generalizing across contexts that share hidden or latent causes^{1–6}. This capacity allows animals to infer the common features between seemingly different experiences and to adjust behavior

accordingly—a process at the core of cognitive mapping^{7,8}. When this ability is disrupted by OFC lesions or inactivation, behavior can become overly tied to superficial idiosyncratic features, leading to inflexible or maladaptive responses when it is necessary to generalize across hidden states to update behavior⁹, as for example in behavioral settings such as after reversal or reinforcer devaluation or in sensory preconditioning^{10–15}.

Notably impairments in such OFC-dependent tasks have also been found to occur in experimental settings after experience with addictive drugs, particularly psychostimulants such as cocaine and amphetamine^{13–24}. Such results, coupled with evidence that addictive drugs affect markers of function in OFC and related areas^{23,25–33}, suggest that drug-induced changes in OFC-dependent processing may underlie particularly pernicious features of substance use disorders, such as craving and relapse, in which maladaptive behaviors return despite treatment and even periods of abstinence^{10,34,35}. One possible explanation for this persistence is that chronic drug exposure compromises the OFC's ability to recognize hidden similarities between situations, thereby disrupting generalization across task states. Combined with pre-existing conditions and other environmental insults in certain individuals, such an effect could lead to the loss of behavioral control that characterizes addiction³⁶.

To investigate this possibility, we examined how prior cocaine use affects OFC representations of hidden states in a sequential decision-making task. Rats were trained to self-administer either cocaine or sucrose and were then recorded from the lateral OFC while performing an odor-based sequence task with positions that either shared or differed in sensory cues. This task design allowed us to assess whether OFC neurons appropriately compressed or generalized across positions with different sensory features but identical behavioral relevance—a hallmark of hidden state identification. As expected, OFC neurons in controls showed near-chance discrimination between comparable positions (P2 and P3) across sequences, reflecting compression of irrelevant sensory differences, and even at positions with unique sensory cues (P1 and P4), neural discrimination was near chance in most trial epochs. This neural compression reflects the OFC's preference to represent latent task states rather than external features. By contrast, cocaine-experienced rats maintained significantly higher selectivity at both shared and unique positions, failing to compress positions that were behaviorally equivalent. Their behavior also became more variable, suggesting reduced recognition of underlying equivalence across sequences. Tensor component analysis (TCA) showed that this loss of generalization extended beyond specific position pairs: cocaine-experienced rats lacked higher-order components that generalized across all positions, indicating a fundamental alteration in how OFC organizes task representations.

Results

To examine the potential impact of cocaine use on the identification of hidden states by OFC neurons, we used a go, no-go odor discrimination task (Figure 1A[↗]). This task, used previously to record in OFC³⁷, featured 6 odors arranged in two 4-odor sequences, labeled as S1 and S2 (Figure 1B[↗]). These sequences had unique odors at the start and end positions (P1 and P4) and shared odors at the two positions in the middle, where the sequences overlapped (P2 and P3). The experiment was conducted over a 4-month period consisting of several phases (Figure 1C[↗]). Initially, naive rats underwent a four-week training on the “figure 8” task. In the subsequent week, rats underwent jugular catheterization surgery and were given time to recover. For the following two weeks, rats were trained to self-administer either sucrose (10% wt/vol; $n = 3$) or cocaine (0.75 mg/kg/infusion; $n = 3$) (Figure 1D[↗]), using procedures similar to those known induce incubation of craving³⁸. The subsequent three weeks were dedicated to electrode implantation surgery in OFC and post-surgical recovery. Two additional weeks were then allocated for reminder training on the odor task. Finally, *in vivo* recordings were obtained during the last four weeks while the rats performed the odor task.

During these recording sessions, rats in both groups displayed consistently high levels of discrimination performance across all positions in the odor sequence (Figure 1F[↗]). In this study, rats were first trained on the odor sequence task to criterion before undergoing cocaine self-administration. Thus, all animals had already acquired the task thoroughly prior to drug exposure,

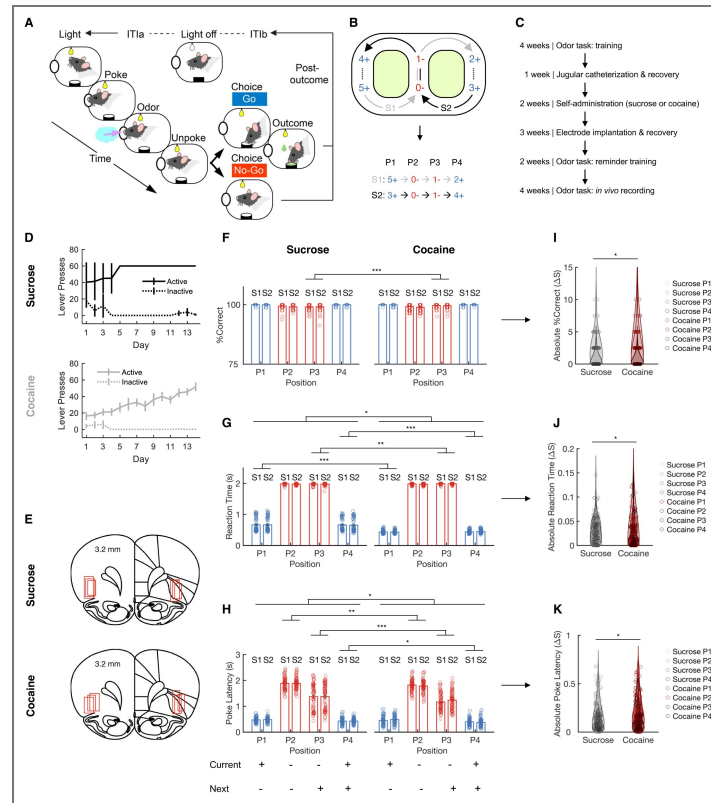


Figure 1. Figure 8 odor sequence task, cocaine self-administration, and behavior

(A) Schematic representation of the trial events in the odor-sequence task. The initiation of each trial was indicated by the illumination of two overhead houselights. After poking into the central odor port and sampling the presented odor, rats had the option to respond with a 'go' to receive a sucrose reward or a 'no-go' to avoid a prolonged inter-trial interval. (B) The six odors were grouped into two sequences, S1 and S2, each consisting of four odor positions (P1–P4). The sequences alternated in a "figure eight" pattern. The numbers at each position represent odor identities, and the blue/red symbols (+/–) indicate rewarded and non-rewarded trials. (C) Timeline of the experimental procedure. (D) The number of active (solid lines) and inactive (dashed lines) lever presses during sucrose self-administration sessions (upper; $n = 3$) and cocaine self-administration sessions (lower; $n = 3$). (E) Reconstruction of recording sites, with red squares indicating the locations of electrodes. (F) Percentage of correct responses (% correct) are shown for each trial type during single-unit recording sessions. A significant group difference emerged specifically at position P3 between the Sucrose and Cocaine groups ($F_{(1,288)} = 13.1$, $p = 3.4 \times 10^{-4}$, $\eta_p^2 = 0.043$; One-way ANOVA). Blue denotes rewarded trial types, while red denotes non-rewarded trial types. (G) Reaction time was defined as the interval between odor port exit and water well entry. For correct no-go trials, where no movement toward the water well was made, a fixed reaction time of 2 seconds—the full response window—was assigned. Reaction time differed significantly between the Sucrose and Cocaine groups overall ($F_{(1,158)} = 6.6$, $p = 0.01$, $\eta_p^2 = 0.006$; One-way ANOVA), with position-specific analyses revealing significant differences at P1, P3, and P4 (P1: $F_{(1,288)} = 152.9$, $p = 1.9 \times 10^{-28}$, $\eta_p^2 = 0.35$; P3: $F_{(1,288)} = 9.3$, $p = 2.5 \times 10^{-3}$, $\eta_p^2 = 0.03$; P4: $F_{(1,288)} = 139.6$, $p = 1.6 \times 10^{-26}$, $\eta_p^2 = 0.33$; One-way ANOVA). (H) Poke latency, measured as the time from light onset to odor port entry. Significant group differences were found across positions ($F_{(1,158)} = 4.0$, $p = 0.04$, $\eta_p^2 = 0.004$; One-way ANOVA), with specific differences emerging at positions P2, P3, and P4 (P2: $F_{(1,288)} = 10.0$, $p = 1.7 \times 10^{-3}$, $\eta_p^2 = 0.034$; P3: $F_{(1,288)} = 12.4$, $p = 5.0 \times 10^{-4}$, $\eta_p^2 = 0.04$; P4: $F_{(1,288)} = 5.7$, $p = 0.018$, $\eta_p^2 = 0.02$; One-way ANOVA). (I) A two-way ANOVA revealed significant main effects of group ($F_{(1,572)} = 4.0$, $p = 0.045$, $\eta_p^2 = 7.0 \times 10^{-3}$) on the absolute difference in percent correct between S1 and S2 for the Sucrose and Cocaine groups. Sucrose data are shown as black color with increasing shading intensity from P1 to P4, while Cocaine data are shown as red color with similarly graded shading from P1 to P4. (J) For the absolute S2–S1 difference in reaction time across positions between Sucrose and Cocaine groups, two-way ANOVA revealed significant main effects of group ($F_{(1,572)} = 6.4$, $p = 0.012$, $\eta_p^2 = 0.01$) and position ($F_{(3,572)} = 3.3$, $p = 0.021$, $\eta_p^2 = 0.017$). Black circles denote sucrose data, with shading that deepens progressively from P1 through P4. Cocaine data are indicated by red circles, with coloration shifting from light to dark red across positions P1 to P4. (K) Two-way ANOVA revealed significant main effects of group ($F_{(1,572)} = 5.6$, $p = 0.018$, $\eta_p^2 = 0.01$) and position ($F_{(3,572)} = 37.7$, $p = 3.2 \times 10^{-22}$, $\eta_p^2 = 0.16$) on the absolute poke latency difference between S1 and S2 in Sucrose and Cocaine groups. Sucrose data are plotted as graded black circles, with shading progressing from light to dark across positions P1 to P4. Cocaine data are plotted as graded red circles, with shading progressing from light red to dark red across positions P1 to P4. Error bars represent standard errors of the mean (SEMs). * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

and the task itself was relatively simple. Nevertheless, we observed several robust and significant differences between the cocaine- and sucrose-trained groups. Differences in trial initiation latencies were observed (Figure 1H [↗](#)), suggesting that the rats in both groups utilized the predictable odor sequence to anticipate the availability of reward not only in the current trial but also in subsequent trials. ANOVAs revealed a variety of small but significant group differences in these measures (Figure 1F-H [↗](#)). In addition to these somewhat idiosyncratic differences, analyses also showed effects in line with the proposal that past cocaine use had affected the rats' ability to ignore external differences between positions across the two sequences. Specifically, the behavior of the rats that had self-administered cocaine was more variable across sequences. This was evident in the distribution of difference scores between positions in each sequence on all three measures, which were larger in the cocaine group than in controls (Figure 1I-K [↗](#)). Thus, even after extensive training, animals with a history of cocaine exposure maintained a stronger behavioral distinction between the same position in the two sequences, in all three behavioral measures.

Prior cocaine use disrupts identification of hidden states across position pairs

During these sessions, we recorded a total of 1,699 units in the lateral OFC of three rats in the cocaine group and 2,182 units from three rats in the control group (Figure 1E [↗](#)). A cursory examination of single-cell examples (Figure 2 A-D [↗](#)) indicated that prior cocaine use was associated with less uniform neural activity at individual positions across sequences. To investigate this effect, we tested whether cocaine use affected the ability of single units and ensembles in the OFC to discriminate between positions in the task, starting with comparisons between the same positions on sequences S1 and S2. Neuronal selectivity on trials at comparable positions in the two sequences was calculated for each trial epoch, and the results showed that the proportion of selective units was very low—essentially at chance during trials at P2 and P3—before increasing at the end of P3 and into the odor sampling periods of P4 and P1. This pattern reflected compression of positions in the central arm of the figure-8 maze, where the two sequences were identical, and discrimination of positions on the outer arms of the maze, where the odors differed. Notably, however, even for P4 and P1, most epochs within the trial showed near chance levels of selectivity in controls, while neurons recorded in cocaine rats maintained significantly higher levels of selectivity (compared both to controls and chance) at the end of P3 and during most of the epochs in P4 when compared directly between the two groups (Figure 2E [↗](#)).

To quantify this effect, we conducted a two-way ANOVA on the activity of each neuron at each position in the two sequences, with sequence and position as factors. This analysis revealed disproportionate effects of cocaine on the prevalence of neurons showing a significant interaction between these two factors (Figure 3—figure supplement 1 [↗](#)), suggesting increased divergence in neural activity at similar positions across sequences in the cocaine group. This impression was reinforced by a second analysis, in which we correlated the preferred position of each neuron in each epoch across sequences. For this, we first identified position selective neurons independently at each epoch and on each sequence (ANOVA, $p < 0.01$). A neuron's preferred position was then taken as the position with the highest firing, and then we calculated a correlation coefficient for each group across all neurons and epochs. As shown in the plot, the Sucrose group exhibited a steeper correlation compared to the Cocaine group. To statistically compare the correlation coefficients, we used Fisher's r -to- z transformation and found a significant difference between the groups (Figure 3—figure supplement 2 [↗](#)).

The increase in differential activity across sequences in the cocaine group was also evident in an ensemble decoding analysis (Figure 3A [↗](#)). Decoding accuracy was above chance for most epochs at positions P1 and P4 and at the end of the trial and into the ITI period at P3, and while this was true in both groups, the decoding accuracy was significantly higher in the cocaine group in nearly all epochs. This difference between groups was even more evident when average decoding across epochs at each position was directly compared between the two groups (Figure 3B [↗](#)). Thus, both

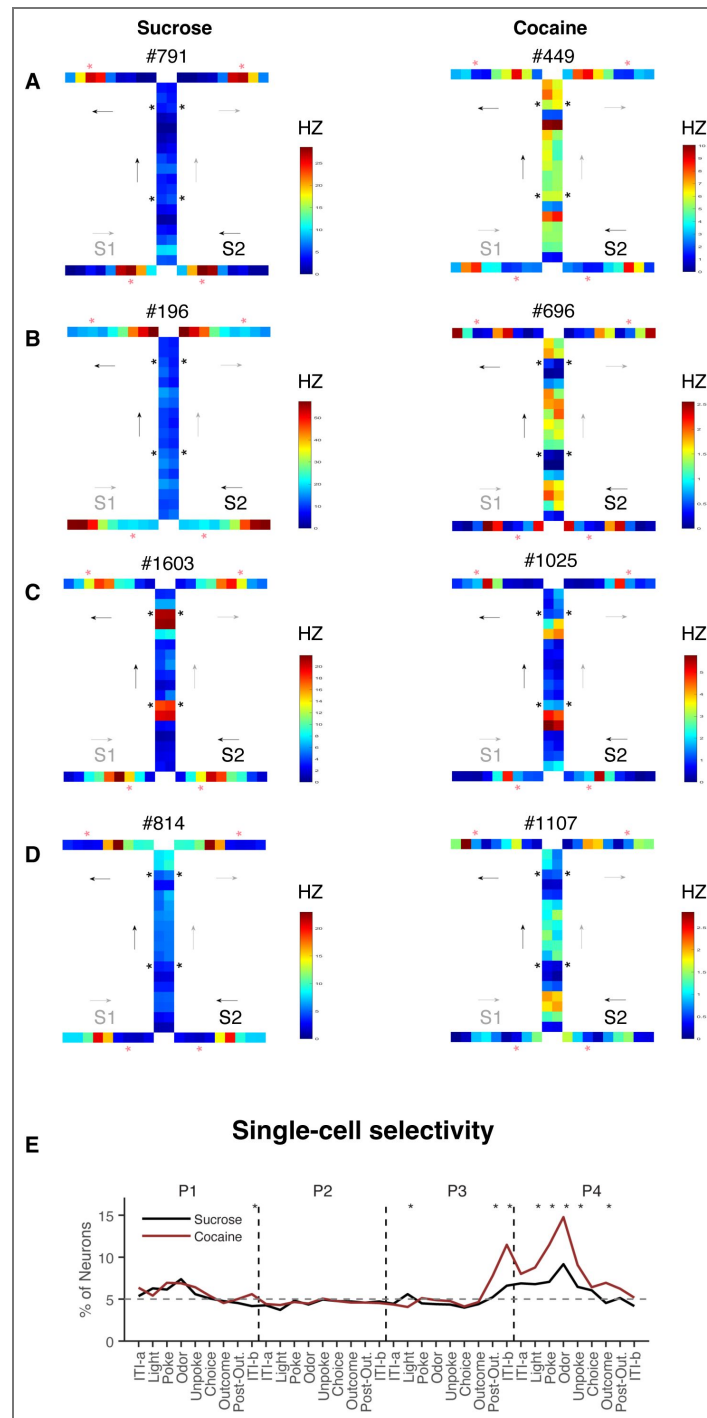


Figure 2. Discriminating sequences S1 versus S2 at single-unit level

(A–D) Heatmaps showing the activity of individual OFC neurons. Each panel illustrates the firing pattern of a single neuron, capturing its response dynamics across task epochs. Each small grid represents one epoch at a specific position, arranged in the direction of the arrow following the sequence: “ITI-a”, “Light”, “Poke”, “Odor”, “Unpoke”, “Choice”, “Outcome”, “Post-Out”, “ITI-b”, repeated for all four positions of each sequence. The left columns represent single OFC neurons from the Sucrose group, where the intensity of left-side epochs closely matches that of the right-side epochs, indicating strong generalization. In contrast, the Cocaine group shows comparatively weaker generalization across sides. S1, sequence 1; S2, sequence 2. (E) Line plot showing neuronal selectivity at each position in S1 versus S2 for each neuron in both the Sucrose (black line) and Cocaine (red line) groups. The selectivity was calculated at different task epochs for all four positions. Each asterisk indicates that there are significant differences between the two groups (χ^2 's > 4.8; p 's < 0.03; Chi-squared test). * p < 0.05; error bars are SEMs.

single unit and ensemble activity in lateral OFC in cocaine-experienced rats compressed meaningful task epochs less than in sucrose-trained controls. Additional analyses examining decoding across all positions within- and across-sequences showed that the Cocaine group exhibited significantly higher decoding accuracy within-sequence and significantly lower decoding accuracy across-sequence, compared to the Sucrose group (Figure 3D-G [↗](#)).

We propose that OFC representations develop through multiple stages, progressing from sensory- and reinforcement-driven coding to sequence-dependent differentiation as animals acquire task structure. With extended training, these representations are normally refined such that functionally equivalent states are compressed, and behaviorally irrelevant distinctions are suppressed. Prior cocaine exposure appears to disrupt this later refinement stage, leaving OFC representations in an earlier, sequence-specific state despite extensive training, consistent with impaired generalization across latent task states. Accordingly, the effect of cocaine was similar to the effect of diminished training on the task. This is evident in a comparison to data from OFC in rats in a prior study using the same task [37](#), in which there was much less training prior to recording (6 weeks of odor task training in total plus 2 weeks of self-administration in this study vs. ~3 weeks of odor task training in the prior study). OFC neurons recorded in these rats revealed preserved decoding relative to the over-trained controls in the current study, decoding that was similar to that in the cocaine rats (Figure 3C [↗](#)). Thus, neural activity in OFC evolves during learning, to identify the underlying hidden states that define behavioral relevance. This refinement or its specificity is disrupted by prior cocaine use.

Prior cocaine use disrupts identification of hidden states across all positions

While the planned analyses conducted above show that prior cocaine use was associated with preserved discrimination of incidental information about task positions normally compressed by rats performing this task, they do not address whether this is a general effect or whether it only impacted direct comparison of the position pairs highlighted by the analyses. That is, did neurons in the cocaine-experienced rats maintain information about these positions because of the idiosyncratic features of our task, or is there a general effect of cocaine on the ability of the OFC to register common features and underlying causes. To get at this more general question, we utilized tensor component analysis (TCA) to identify the dominant factors shaping neural activity across units, epochs, and trials in our task. This analysis provides a less constrained, more hypothesis-agnostic approach to our question since it looks for patterns across all of these factors in the design. If cocaine is causing a failure of generalization more broadly, then we would expect loss of power in dimensions across positions and not just within position pairs.

The results for this analysis are shown in Figures 4A-B [↗](#) and Figure 4—figure supplement 1 [↗](#). Neural variance was partitioned into components, one per row. The analysis shown was constrained to 10 components, as this number produced the most consistent and reliable results compared to other choices (Figure 4—figure supplement 2 [↗](#)-3 [↗](#)); additionally, the features we will highlight for each group were generally consistent within individual rats from each group (Figure 4—figure supplement 4 [↗](#)-5 [↗](#)). For each group, the characteristics of each component are visualized by the contributing neuron weight (left), temporal dynamics within a trial (middle), and amplitude of such dynamics across trials organized by positions (right). The temporal factor captures the activity pattern across epochs within a trial, whereas the trial factor reflects how strongly that pattern is expressed across positions and trials. Because TCA components are scale-indeterminate, factor magnitudes are meaningful only relative to one another within a component, not across components. Thus, differences in trial factors with similar temporal dynamics indicate differential recruitment of the same within-trial activity pattern across task positions, rather than changes in response timing. To discuss one example, the top row in the control group shows that the first component was broadly distributed across neurons (left panel), expressed most strongly at epochs of Unpoke, Choice and Outcome (middle panel), and exhibited high amplitude at P1 and P4 and low at P2 and P3 (right panel). Further, it was higher in P3 than P2 and also showed similar patterns across trials within each of these position pairs (right panel).

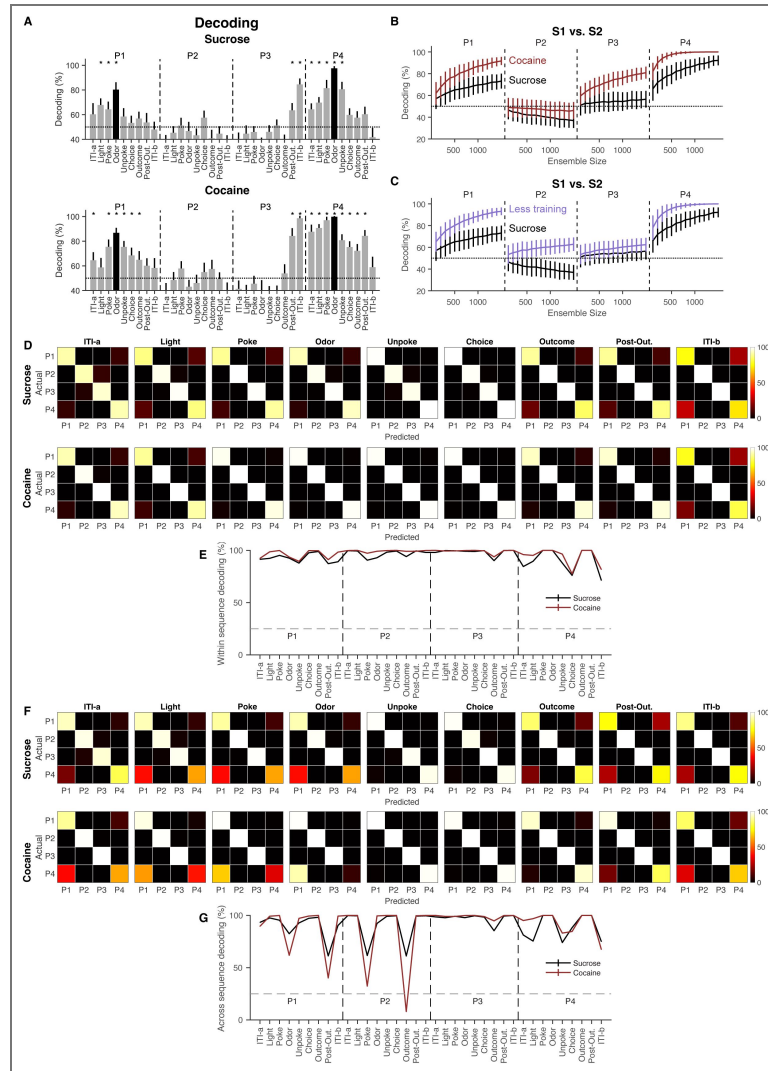


Figure 3. Cocaine use reduces the ability of OFC to generalize task-irrelevant odor-overlapping sequences.

(A) Decoding accuracy of S1 versus S2 was evaluated within each of the nine task epochs for positions P1 to P4. Error bars represent SDs, and each asterisk indicates that the mean decoding accuracy exceeds a 95% confidence interval calculated using the same decoding process with label-shuffled data. The dotted lines represent the chance level of decoding. The meaning of the black bars is consistent with Figure 2E. (B) Decoding accuracy of S1 versus S2 at each position was assessed using varying ensemble sizes for the Sucrose (black) and Cocaine (red) groups across all epochs. The Cocaine group demonstrated higher decoding accuracy and reduced generalization of task-irrelevant, overlapping sequences compared to the Sucrose group ($F_{(1,118)} = 16.8$, $p = 7.8 \times 10^{-5}$, $\eta_p^2 = 0.12$; One-way ANOVA). P1, P2, P3, and P4 represent positions 1, 2, 3, and 4, respectively. (C) Decoding accuracy of S1 versus S2 at each position was assessed using varying ensemble sizes for the Sucrose group (black, same data as in Figure 3B) and the Less Training group (blue, Zhou et al., 2021) across all epochs. The decoding accuracy in the Less Training group was comparable to that observed in the Cocaine group (Figure 3B, red) ($F_{(1,118)} = 0.01$, $p = 0.92$, $\eta_p^2 = 9.0 \times 10^{-5}$; One-way ANOVA). Additionally, the Less Training group exhibited significantly higher decoding accuracy than the Sucrose group ($F_{(1,118)} = 18.1$, $p = 4.2 \times 10^{-5}$, $\eta_p^2 = 0.13$; One-way ANOVA). P1, P2, P3, and P4 represent positions 1, 2, 3, and 4, respectively. * $p < 0.05$; error bars indicate SDs. (D) Confusion matrices showing within-sequence position decoding from OFC ensemble activity at four task positions for the Sucrose and Cocaine groups. The y-axis denotes the rats' actual position, and the x-axis indicates the predicted position. Brighter colors reflect higher decoding probabilities. (E) Quantification of within-sequence decoding accuracy across positions revealed a significant difference between groups ($p = 0.027$; $W = 1118$; two-sided Wilcoxon rank-sum test), with Sucrose shown in black and Cocaine in red. (F) Confusion matrices showing across-sequence position decoding from OFC ensemble activity at the same four task positions. Axes are as in (D), with brighter colors indicating higher decoding probabilities. (G) Quantification of across-sequence decoding accuracy also revealed a significant group difference ($p = 0.046$; $W = 1144$; two-sided Wilcoxon rank-sum test), with Sucrose shown in black and Cocaine in red.

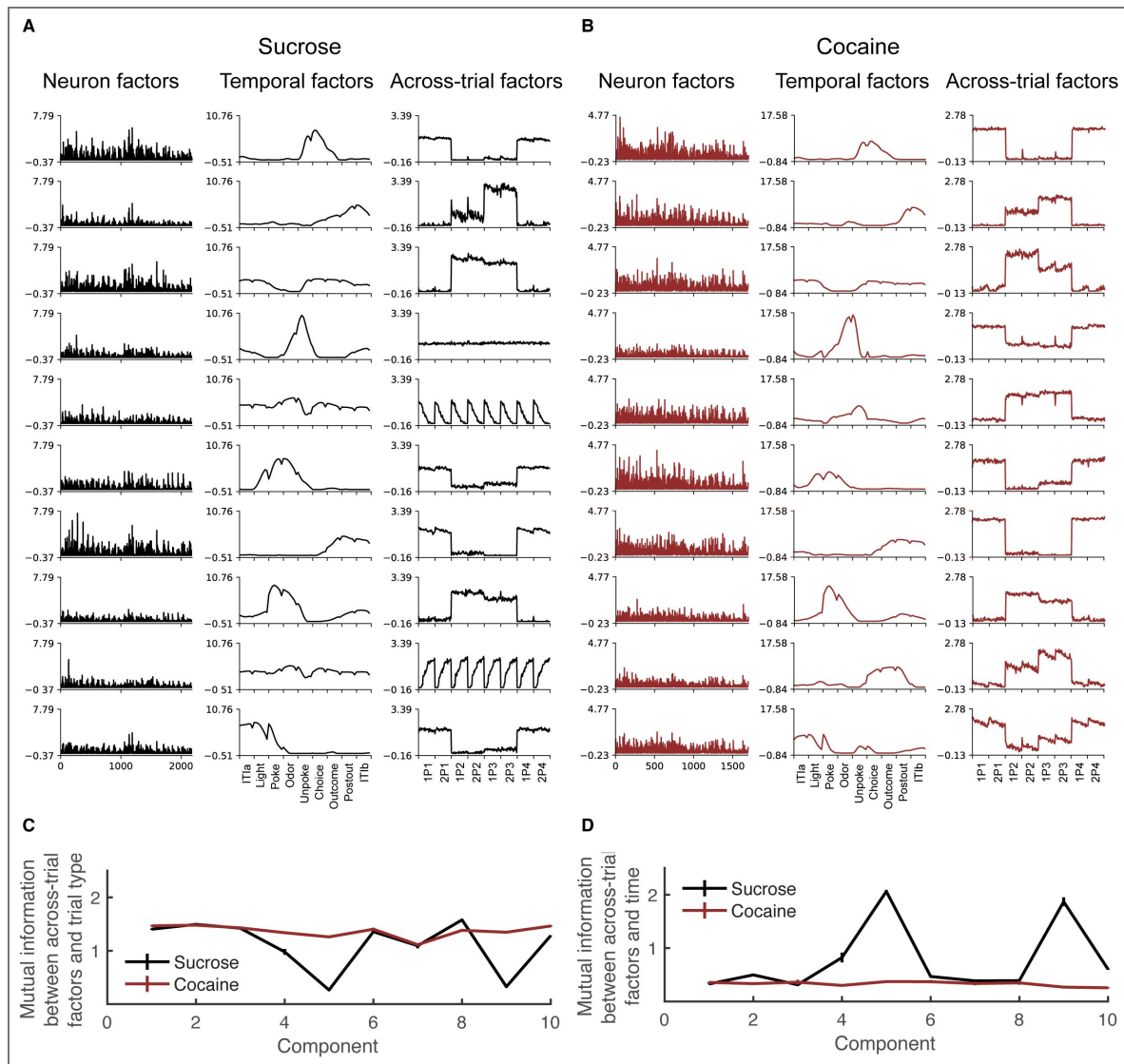


Figure 4. Cocaine use impairs the OFC's function in temporal cognition and decreases the generalization of task-irrelevant odor-overlapping sequences

(A–B) Ten components of TCA applied to Sucrose(A) and Cocaine(B) neuron activity are displayed. Each component consists of a neuron factor (left column), a temporal factor spanning nine events with eight time points per event (middle column), and a trial factor (right column). Temporal factors are grouped as follows: 1–8 (preTrial), 9–16 (Light), 17–24 (Poke), 25–32 (Odor), 33–40 (Unpoke), 41–48 (Choice), 49–56 (Outcome), 57–64 (postTrial1), and 65–72 (postTrial2). The dataset includes eight trial types, each with 40 trials. Trial factors 1–80 and 241–320 represent rewarded (positive) trials, while 81–240 correspond to unrewarded (negative) trials. Trials are further categorized as follows: 1–80 (P1), 81–160 (P2), 161–240 (P3), and 241–320 (P4). These low-dimensional components were extracted from a 10-component model for both the Sucrose(A) and Cocaine groups(B). In the Sucrose group, components 1, 6, 7, and 10 encode positive trials, while components 2, 3, and 8 encode negative trials. Similarly, in the Cocaine group, components 1, 4, 6, 7, and 10 encode positive trials, whereas components 2, 3, 5, 8, and 9 encode negative trials. Notably, in the Sucrose group, beyond reward-related components, additional components encode aspects of temporal cognition (components 5 and 9), along with early components broadly active across most trials (component 4). (C) Plot illustrating mutual information (MI) between across-trial factors and trial types. The Sucrose group exhibits lower MI, particularly at components 5 and 9. A two-way ANOVA confirmed a significant reduction in MI for the Sucrose group ($F_{(1,1360)} = 571.6$, $p = 9.7 \times 10^{-106}$, $\eta_p^2 = 0.3$), especially at these components, which showed consistent patterns across all trial types, aligning with the plots in (A) of the Sucrose group, indicating a weaker dependency between across-trial factors and trial types. A significant difference was also observed across components ($F_{(9,1360)} = 266.3$, $p = 1.2 \times 10^{-292}$, $\eta_p^2 = 0.64$). (D) Plot illustrating mutual information (MI) between across-trial factors and time. The Sucrose group demonstrates greater MI, particularly at components 5 and 9, suggesting a stronger dependency between across-trial factors and time. A two-way ANOVA confirmed a significant increase in MI for the Sucrose group ($F_{(1,1360)} = 850.1$, $p = 1.4 \times 10^{-145}$, $\eta_p^2 = 0.38$). Additionally, a significant difference was observed across components ($F_{(9,1360)} = 176.8$, $p = 1.4 \times 10^{-221}$, $\eta_p^2 = 0.54$). Error bars are SEM.

After aligning these component patterns between groups based on their temporal factors, we observed different patterns in trial amplitudes reflecting differences in positional representations (right columns). In controls, most of the trial factors distinguished between the position pairs (rows 1–3, 6, 7, 8, 10). These factors distinguished rewarded from non-rewarded position pairs, in some cases further distinguishing particular position pairs within each category (rows 2,3,7,8) and/or showing effects of trials (rows 2,3). However, none showed differences between positions or trials inside a pair. This result is consistent with findings in [Figure 2](#) and [3](#), which showed that both individual units and ensemble responses in control rats collapsed the positions within each pair. However, controls also had three additional factors that appeared to generalize not just across positions inside a pair but rather across all positions (rows 4, 5, 9). These factors were identical across all 8 positions and two (rows 5 and 9) showed identical changes across trials within each position. This indicates a significant tendency in neural activity in OFC to represent commonalities across all 8 positions dynamically across time in a session; this global generalization goes beyond that illustrated in our planned comparisons in [Figures 2](#) and [3](#).

The trial factor patterns in the cocaine-experienced rats were substantially different; all of the 10 factors distinguished rewarded versus non-rewarded position pairs, and the position-general factors evident in controls (rows 4, 5, 9) were absent from activity in the cocaine rats. This dichotomy is consistent with reduced compression evident in our planned comparisons in [Figures 2](#) and [3](#) and further points to diminished generalization across task features and a heightened emphasis on differences across positions in the task, which presumably reflects the different values imparted by the sequences. These general features were robust and interpretable across animals in both the Sucrose and cocaine-exposed groups ([Figure 4—figure supplement 4–5](#)). This consistency supports the validity of comparing TCA-derived measures across groups, and consistency across individuals within each group.

To quantify the differences in [Figures 4A–B](#), we used mutual information (MI) to measure the information available in each factor—particularly factors 5 and 9 about position and trial. Consistent with the description above, we found lower MI values for position in controls at these components compared to the cocaine group ([Figure 4C](#)) and higher MI at the same components for trial or temporal information ([Figure 4D](#)). Moreover, similar results were observed across varying numbers of components, with the significant difference between the two groups remaining consistent regardless of the number of components selected ([Figure 4—figure supplement 2–3](#)).

Discussion

Prior to considering the implications of the current results, several limitations should be noted. First, sucrose self-administration is not a neutral manipulation and may itself promote abstraction in OFC representations by providing additional instrumental experience with the same reinforcer used in the odor-guided task. Second, the number of animals per group was relatively small, and all subjects were male, limiting the ability to fully assess individual variability, potential sex-dependent effects, or differences in motivational state or attentional engagement that may have contributed to the observed effects. Nonetheless, the key neural and behavioral signatures were consistent across individuals and analytic approaches, with no outliers observed, and sample sizes of this scale are common in cocaine self-administration studies due to their technical and logistical demands.

These results are consistent with a cocaine-induced disruption of the normal identification of underlying hidden states by OFC^{1–6}. Specifically, in the current task, OFC representations normally evolve with training to compress or generalize similar positions in the two sequences, even when external sensory information—the odor cues—differ. Cocaine-experienced rats failed to show this normal compression of irrelevant information, instead discriminating these position pairs at higher rates, similar to encoding observed in rats early in learning. Additionally, this failure to generalize was also evident in a TCA analysis, where factors generalizing across different positions were prominent in control data but entirely absent in data from cocaine-experienced rats.

The loss of this normal function of OFC is relevant to addiction, since it suggests that some addictive drugs, at least the psychostimulants, cause fundamental and long-lasting changes in how prefrontal areas process task-related information. These effects are consistent with prior studies showing changes in OFC function after drug use, particularly for behaviors in which it is necessary to generalize across hidden task states to appropriately update behavior when likely outcomes change, such as after devaluation and in sensory preconditioning or even reversal ^{10–15}. In rodents, drug-induced changes in OFC-dependent learning and insight are accompanied by long-lasting alterations in OFC neural activity, including degraded single-unit encoding of expected outcomes, altered ensemble representations of task structure, and persistent changes in other properties of OFC neurons ^{15,22,23,27,28,30}. Convergent findings in nonhuman primates and humans further demonstrate that drug exposure is associated with OFC dysfunction and inflexible choice behavior ^{13,16,24,39}, supporting a conserved role for OFC across species in guiding adaptive decision making. Together, this body of work also suggests that cocaine self-administration induces enduring changes in OFC computations that undermine generalization across hidden task states, providing a mechanistic link between altered neural coding in OFC and the behavioral inflexibility that characterizes addiction.

Consistent with such speculation, the current findings show a specific effect of cocaine use on generalization or the ability of OFC neurons to collapse trivial external information to encode underlying causes that different situations have in common ³⁶. A loss of this ability could have wide ranging effects ⁴⁰, but it might particularly disrupt the mobilization of learning from other settings to counteract or diminish drug-seeking behaviors. For instance, consequences of drug use learnt in separate contexts or situations – for instance at home or in the classroom, during counseling, or even from observing the impact of drug use in the lives of others – would not be as effectively deployed to affect behavior during one’s own drug-seeking. Similarly therapeutic approaches designed to extinguish drug-seeking would also generalize more poorly outside of the clinical setting. The present results identify a neurophysiological mechanism—loss of OFC-mediated generalization—that may underlie the persistence and context-specificity of drug-seeking behavior, providing a new window into how addictive drugs alter cognitive mapping and flexible decision-making in the brain.

Methods

Contact for reagent and resource sharing

This study did not generate any unique reagents. However, for further information or requests for resources and reagents, please contact the Lead Contact, Geoffrey Schoenbaum (geoffrey.schoenbaum@nih.gov .

Experimental model and subject details

The study was conducted on a group of six male Long-Evans rats (Charles River strain), aged around three months and weighing between 175–200 g. The rats were housed individually in an AAALAC-accredited animal care facility at the National Institute on Drug Abuse Intramural Research Program (NIDA-IRP), with *ad libitum* access to food on a 12-hour light-dark cycle. Water was removed a day before testing, and the rats were provided with free access to water for 10 minutes each afternoon in their home cages. If there was no testing scheduled for the following day, they were given free access to water. All behavioral testing was conducted at the NIDA-IRP, and the animal care and experimental procedures were conducted in accordance with the guidelines set by the US National Institutes of Health (NIH) and approved by the Animal Care and Use Committee (ACUC) at the NIDA-IRP.

Method details

Behavioral Tasks

Figure 8 Task

The study employed aluminum boxes (18 in. on a side) equipped with a port for odor delivery and a well for delivery of sucrose solution for conducting behavioral training. A custom-written C++ program and a system of relays and solenoid valves were used to control the task events. The entries into the odor port and the fluid well were detected by infrared beam sensors. The availability of each trial was signaled by the illumination of two house lights above the odor port. The trial was initiated if the rat entered the odor port within 5 s after light onset, leading to odor delivery after a 500-ms delay. The rats were required to remain in the port for an additional 500 ms; otherwise, the trial was aborted, and the lights extinguished. After 500 ms, the rats were free to leave the port, terminating odor delivery. Post port exit, the rats had 2 s to respond at the fluid well. Responding on rewarded trials led to the delivery of a sucrose solution (10% w/v; 50 μ L) after a random delay ranging from 400 to 1,500 ms. On nonrewarded trials, nonresponding during the 2-s period, or responding after the 2 s, the house lights were extinguished, indicating the end of the trial and the beginning of the ITI. Correct trials were followed by a 4-s ITI, and trials on which the rat made an error were followed by an 8-s ITI.

The study for the Figure 8 task included six odors, organized into two sequences (S1 and S2) that occurred repeatedly in turn (S1 $\rightarrow$ S2 $\rightarrow$ S1 $\rightarrow$ S2 $\rightarrow$... $\rightarrow$ S1 $\rightarrow$ S2; 40 repeats of each sequence). On each trial, one of six odors was delivered to the odor port. The odor identity is indicated by a number, and reward and nonrewarded are indicated by the positive (+) and negative (–) symbols, orders of the odors were organized and showed below:

S1: 5+, 0–, 1–, 2+

S2: 3+, 0–, 1–, 4+

To avoid bias, the starting sequence for each session—either S1 or S2—was determined in a fully pseudorandom manner. Before training with any odors, rats were first shaped to nosepoke at the odor port and then respond at the well for a reward. The rats were trained on the full set of sequences from Day 1 until they achieved >75% accuracy on every trial type in a session. Following this, electrode arrays were implanted bilaterally in the orbitofrontal cortex.

To clarify, the animals were not initially trained on the “Figure 8” task (which involves six odors across two four-odor sequences, S1 and S2) prior to exposure to the full 24-position task. Each rat in this study was trained and recorded on only one behavioral paradigm—either the Figure 8 task or the 24-Position Odor Sequence Task, but not both. These tasks were conducted in separate cohorts of animals. All data presented in the figures of this manuscript were obtained from the Figure 8 task, using sucrose-trained, cocaine-trained, or previously reported minimally trained animals, with the exception of Figure 3—figure supplement 1 [\[link\]](#). The data shown in Figure 3—figure supplement 1 [\[link\]](#), derived from the 24-Position Odor Sequence Task, were published previously, and the behavioral paradigm is described in the corresponding figure legend.

Surgical Procedures

Rats were surgically implanted with a total of 32 electrodes, organized into two bundles of 16 electrodes each. These bundles were constructed using nickel-chromium wires with a bare diameter of 25 μ m (AM Systems, WA). The implantation targeted the bilateral orbitofrontal cortices (AP: 3 mm, ML: 3.2 mm). To ensure proper placement, each wire bundle was encased in a 27-gauge stainless-steel tubing and trimmed using fine spring scissors. The trimmed wires extended approximately 1.5–2 mm beyond the tubing’s end. Initially, the wire tips were positioned 4.2 mm ventral from the brain surface. Following the surgical procedure, the rats received oral doses of Cephalexin (15 mg/kg) twice daily for a duration of two weeks to prevent any potential infections.

Catheter Surgery

Rats used for cocaine self-administration received chronic indwelling jugular catheter implants (Instech Laboratories). Rats were anaesthetized using ketamine (100 mg/kg, i.p., Sigma) and xylazine (10 mg/kg, i.p., Sigma). Blunt dissection was performed to isolate right external jugular veins and catheters were surgically implanted 3 cm into the veins. Catheters were passed subcutaneously to the back, where they were attached to an external harness. Carprofen (5 mg/kg,

s.c., Pfizer) was administered after surgery as an analgesic. Rats recovered for 7 days before self-administration began. During recovery and self-administration, catheters were flushed daily with a cocktail of enrofloxacin (4.0 mg/mL, Bayer) and heparinized saline (50 IU/mL in 0.9% sterile saline, Sigma) to maintain catheter patency.

Self-Administration

Following recovery from catheterization surgery, rats were trained to self-administer cocaine-HCl (0.75 mg/kg/infusion; $n = 3$) or sucrose (10% wt/vol; $n = 3$) for 14 consecutive days. Rats were trained in modular behavioral test chambers (Coulbourn Instruments) housed in sound-attenuating boxes. Each chamber was equipped with two levers positioned 8 cm above the floor on opposite sides of the same wall. For intravenous cocaine self-administration, catheter ports were attached to silastic tubing connected to infusion pumps (Med Associates Inc.) located outside sound-attenuating boxes. For sucrose self-administration, sucrose solution was delivered via photobeam-monitored recessed dippers. Daily sessions were 3 hours and began with the illumination of a house light and the insertion of an active lever. Under a fixed ratio 1 (FR1) schedule of reinforcement, active lever presses resulted in 4 s infusions or dipper insertions (0.05 mL), for cocaine-HCl or sucrose, respectively, and were paired with the illumination of a cue light above the active lever. Infusions and dipper insertions were followed by a 40 s timeout period when the active lever retracted and the house light was extinguished. Following the timeout period, the lever was reinserted and the house light was turned back on. Inactive lever presses had no programmed consequence. Reinforcers were limited to 20 *per* hour to prevent overdose in cocaine self-administering rats. When 20 reinforcers were earned in less than an hour, a timeout period as described above was imposed until the beginning of the next hour.

Single-Unit Recording

Electrophysiological signals were recorded using Plexon OmniPlex systems (Plexon, Dallas, TX). These signals were digitized, amplified, and subjected to bandpass filtering (250–8,000 Hz) to isolate spike activity. Manual thresholding was performed on each active channel to capture unsorted spikes. Timestamps for behavioral events were synchronized with the Plexon system and recorded together with the neural activity. To remove noise and identify single units, spike sorting was carried out offline using Offline Sorter (v.4.0; Plexon), utilizing a template-matching algorithm. The sorted files were then processed in NeuroExplorer (Nex Technologies, Colorado Springs, CO) to extract timestamps for both unit activity and behavioral events. Subsequently, these timestamps were exported as MATLAB (2021b; MathWorks) formatted files for further analysis. It is important to note that the electrodes were not advanced within a specific problem. However, we cannot make any claims regarding the consistency of single units recorded on different days within the same problem, as they may represent distinct neurons. To sample different neural populations during odor problems, the electrodes were advanced by approximately 120 μm . Both in vivo recordings and spike sorting were performed in a blinded manner, without knowledge of whether the subject belonged to the Sucrose or Cocaine group.

Quantification and statistical analyses

Quantification and statistical analyses were conducted using MATLAB (R2024b; MathWorks) and Python Software Foundation, 2024. The sample sizes of rats and neurons were not predetermined through specific statistical methods; however, they are consistent with those reported in previous studies conducted by our lab and other research groups.

Task Events and Peri-Event Spike Train Analysis

The trials were divided into nine distinct epochs, each corresponding to different task events: “ITI-a”, “Light”, “Poke”, “Odor”, “Unpoke”, “Choice”, “Outcome”, “postOutcome”, and “ITI-b”. “ITI-a” represented the time point 0.7 seconds before the house-light turned on. On reward trials, the well-entry moment was labeled “Choice.” The “Outcome” epoch denoted the time of reward delivery. On non-reward trials, the end of the 2-second response window was marked as “Choice,” and a time point 0.7 seconds after “Choice” was labeled as “Outcome.” Both on reward and non-reward trials, 0.7 seconds after the outcome was recorded as “postOutcome,” followed by another

0.7 seconds designated as “ITI-b.” Behavioral performance was evaluated by calculating the percentage of trials in which the rats responded correctly and determining the latency at which they initiated a trial after the onset of the light. The spike train for each isolated single unit was aligned to the onset of each task event to create a peri-event time histogram (PETH). The PETH was constructed with a pre-event time of 200 ms and a post-event time of 600 ms, counting the number of spikes within each 100 ms bin. To smooth the PETH on each trial, a Gaussian kernel with a σ (standard deviation) of 50 ms was applied. For further analysis, a random selection of 30 correct trials was made from each trial type, resulting in a total of 240 trials (30 trials $\times$ 8 trial types). The post-event firing rates (100–600 ms) were averaged to obtain a single measure of neural activity for each neuron on each trial during each task epoch.

Classification Analyses

The neural data collected during each task epoch were organized into a two-dimensional matrix, where the rows represented individual trials and the columns represented the firing rates of each neuron across all trials. In other words, each trial was represented as a vector, with each dimension corresponding to the firing rate of a specific neuron. Neurons recorded across different sessions were concatenated, aligning them with the corresponding trials to create pseudoensembles. To remove temporal correlations between neurons and generate different pseudoensembles, we shuffled the trial orders within each trial type. This shuffling process was repeated 10,000 times, resulting in 10,000 pseudoensembles. By using the linear Support Vector Machine (SVM) for classification analyses, we assessed the classification accuracy through a leave-one-out cross-validation procedure. Specifically, one trial from each trial type was excluded for future testing, while the remaining trials were used to train the classifier. For each pseudoensemble, the leave-one-out cross-validation was repeated 200 times to estimate the mean decoding accuracy. The decoding analyses were conducted on the 10,000 pseudoensembles to calculate an overall mean decoding accuracy. To determine the statistical significance of the overall mean decoding accuracy, we estimated a 95% confidence interval by running the same decoding process with label-shuffled pseudoensembles.

Cross-sequence decoding

To assess population decoding of position within and across sequences in OFC cells ($n = 1000$), we applied a Support Vector Machine (SVM) classifier. Decoding accuracy was estimated using a leave-one-out cross-validation approach. In each iteration, 30 trials per trial type were randomly sampled for each of the 9 task epochs, producing a 120 (trials) $\times$ 9 (epochs) matrix per sequence. One trial from each trial type in sequence 1 was withheld for testing, while the trial with the corresponding index in sequence 2 was simultaneously set aside for across-sequence evaluation. The classifier was trained on the remaining sequence 1 trials. For each epoch and trial type, decoding accuracy was averaged across 1000 iterations to obtain the mean performance.

TCA Analysis

To perform TCA, neuronal firing rates for each group were structured into a three-dimensional array ($N \times T \times K$), where N represents the number of neurons, T the time samples *per* trial, and K the number of experimental trials. This array, referred to as a third-order tensor, captures neuronal activity across trials. Data were exported as a MATLAB .mat file and imported into Spyder for analysis using the TensorTools package⁴¹. Temporal factors were grouped as follows: factors 1–8 (preTrial), 9–16 (Light), 17–24 (Poke), 25–32 (Odor), 33–40 (Unpoke), 41–48 (Choice), 49–56 (Outcome), 57–64 (postTrial1), and 65–72 (postTrial2). There are 8 trial types, each with 40 trials. Trial factors 1–80 and 241–320 represent positive trials with reward, while 81–240 correspond to negative trials without reward. Trials 1–80 map to P1, 81–160 to P2, 161–240 to P3, and 241–320 to P4. Error and similarity plots were generated to assess the stability of TCA’s optimization landscape. Specifically, we ran the TCA optimization algorithm 100 times for each of 10 components, initializing each run with random conditions, and plotted the normalized reconstruction error across all iterations. This approach allowed us to evaluate whether certain runs converged to local minima with high reconstruction errors. Additionally, similarity scores were computed for each model relative to the best-fit model with the same number of

components, with the lines representing the mean similarity as a function of component count. Across all tested component numbers, the 100 repeated runs exhibited substantial overlap and consistently produced similarity scores above 0.8, demonstrating high quantitative consistency. MI was calculated for across-trial factors and trial type, as well as across-trial factors and time, after aligning components between the Sucrose and Cocaine groups. The alignment followed the same ordering of across-temporal factors between the two groups. MI was calculated for across-trial factors and trial type, as well as across-trial factors and time, after aligning components between the Sucrose and Cocaine groups. The alignment followed the same ordering of across-temporal factors between the two groups.

Figure supplements

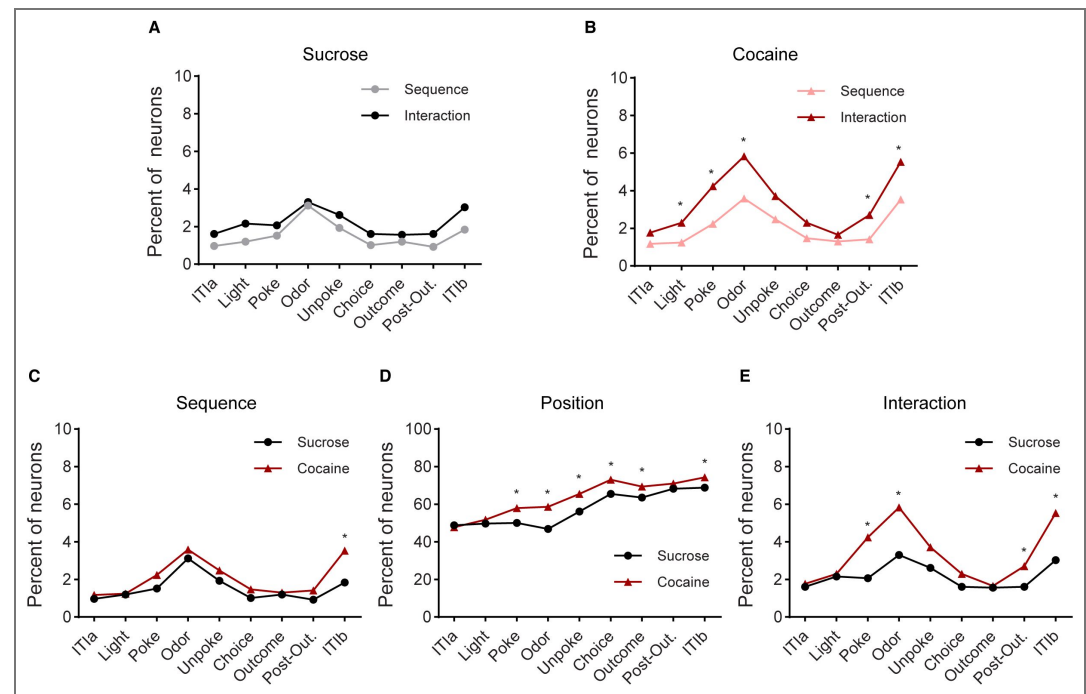


Figure 3—figure supplement 1. Two-way ANOVA of sequence and position effects (A–B) Percentage of neurons showing significant effects of sequence or interaction in the Sucrose (A) and Cocaine (B) groups. No significant sequence or interaction effects were detected across any epoch in the Sucrose group ($\chi^2 < 6.5$, $p > 0.06$; Chi-squared test). In contrast, the Cocaine group showed a markedly higher proportion of neurons exhibiting interaction effects than sequence effects during the Light, Poke, Odor, Post-Out, and ITIb epochs ($\chi^2 > 5.5$, $p < 0.034$; Chi-squared test). (C) Percentage of neurons showing sequence effects. A significant group difference was observed only at the ITIb epoch ($\chi^2 = 11$, $p = 8.3 \times 10^{-3}$; Chi-squared test). (D) Percentage of neurons showing position effects. Significant group differences were found at the epochs of Poke, Odor, Unpoke, Choice, Outcome, and ITIb ($\chi^2 > 13.8$, $p < 3.0 \times 10^{-4}$; Chi-squared test). (E) Percentage of neurons showing interaction effects. The Cocaine group showed significantly more interaction effects than the Sucrose group at epochs of Poke, Odor, Post-Out, and ITIb ($\chi^2 > 14.5$, $p < 4.2 \times 10^{-4}$; Chi-squared test).

Figure 3—figure supplement 2. Correlation of preferred positions

Plots display regression lines for the Sucrose and Cocaine groups. The black line represents the Sucrose group, while the dark red line corresponds to the Cocaine group. A statistical comparison using Fisher's *r*-to-*z* transformation revealed a significant difference between the two groups ($z = 4.92$, $p = 8.6 \times 10^{-7}$). P1, P2, P3, and P4 denote different positions.

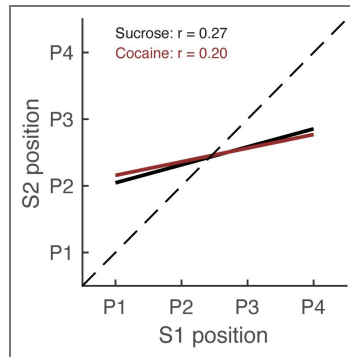
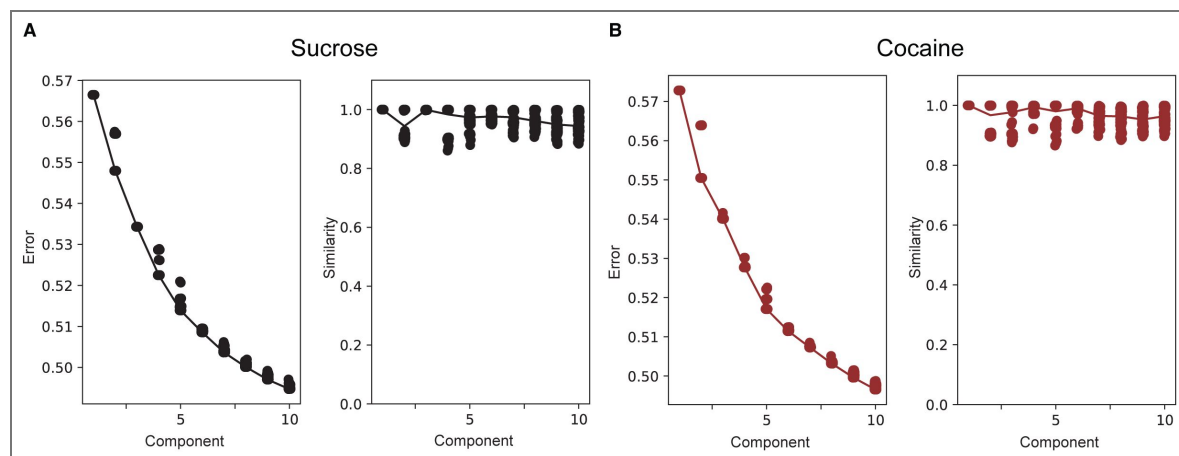


Figure 4—figure supplement 1. Error and similarity plots for TCA

The optimization landscape of TCA may contain suboptimal solutions (local minima), necessitating iterative optimization to minimize a cost function. To evaluate the stability of this process, we generated error and similarity plots. Specifically, we ran the TCA optimization algorithm 100 times for 10 components, each initialized with random conditions, and plotted the normalized reconstruction error across all runs. This approach enabled us to assess whether certain runs converged to local minima with high reconstruction errors. As shown in Figure 4—figure supplement 1A-B (left panels), the error plots indicate that all 100 runs at a fixed number of components produced nearly identical reconstruction errors. Additionally, models with 9 and 10 components exhibited very similar errors. The similarity plot (Figure 4—figure supplement 1A-B, right panel) further quantifies the consistency of TCA models across different component numbers (horizontal axis). For each model, similarity scores were computed relative to the best-fit model with the same number of components, with the lines representing the mean similarity as a function of the number of components. Across all tested component numbers, the 100 repeated runs per component showed substantial overlap and consistently yielded similarity scores above 0.8, indicating high quantitative consistency. Our tests indicate that adding more than ten components reduces model reliability, making them less identifiable. (A-B) Error plots (left panel) and similarity plots (right panel) for Sucrose (A) and Cocaine group (B). Error plots illustrate the normalized reconstruction error across TCA models with varying component numbers. Each black dot corresponds to a model fit using different initial parameters, all of which produced nearly identical performance. Notably, reconstruction error showed almost no further improvement beyond ten components for both groups. The similarity plot displays the similarity score for TCA models with different component numbers. Each black dot represents the similarity of a given model to the best-fit model with the same number of components, demonstrating high repeatability for both groups.



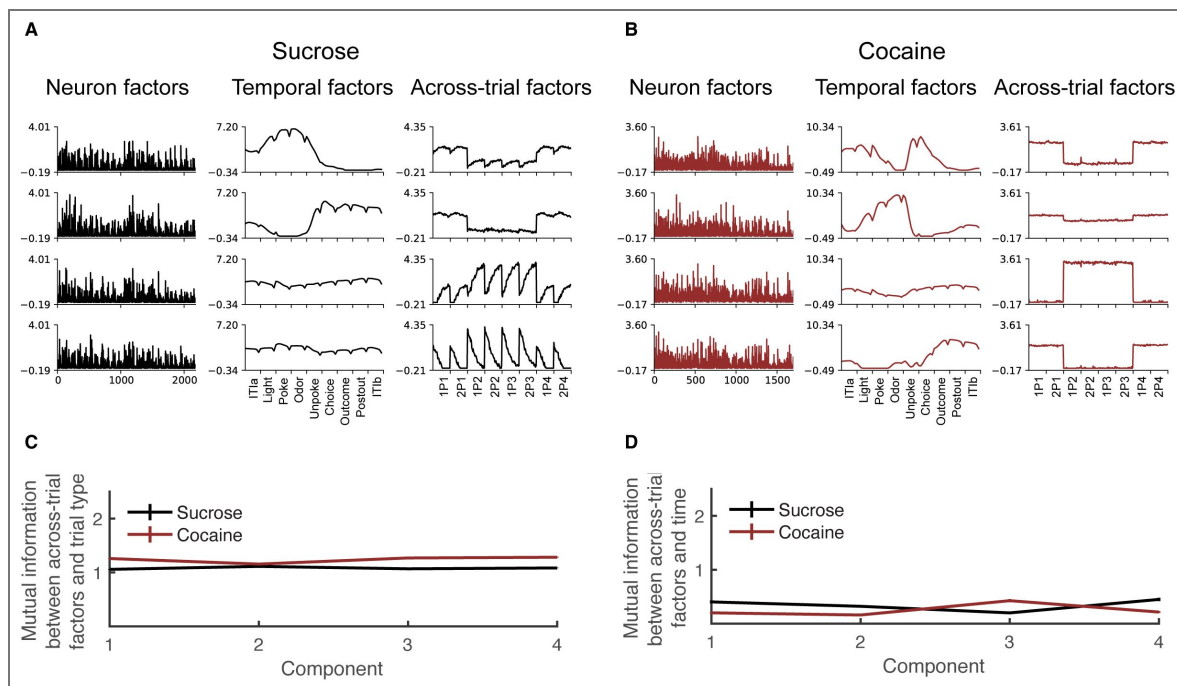


Figure 4—figure supplement 2. Four component model, cocaine impairs OFC temporal cognition and sequence generalization with irrelevant overlapping odors

(A–B) Four components from TCA applied to Sucrose (A) and Cocaine (B) neuron activity are shown. Each component includes a neuron factor (left), a temporal factor spanning 72 time points across nine events (middle), and a trial factor (right). Temporal events are grouped as: 1–8 (preTrial), 9–16 (Light), 17–24 (Poke), 25–32 (Odor), 33–40 (Unpoke), 41–48 (Choice), 49–56 (Outcome), 57–64 (postTrial1), and 65–72 (postTrial2). The dataset contains eight trial types (40 trials each): trials 1–80 and 241–320 are rewarded (P1, P4), and 81–240 are unrewarded (P2, P3). In Sucrose, components 1, 3 and 4 relate to temporal cognition. (C) Mutual information (MI) between across-trial factors and trial types is lower in the Sucrose group, especially at components 1, 3 and 4, indicating weaker dependency. Two-way ANOVA confirmed a significant group difference ($F_{(1,544)} = 1074.5$, $p = 6.7 \times 10^{-131}$, $\eta_p^2 = 0.66$) and component effect ($F_{(3,544)} = 17.7$, $p = 5.2 \times 10^{-11}$, $\eta_p^2 = 0.089$). (D) MI between across-trial factors and time is higher in Sucrose, particularly at components 1, 2 and 4, reflecting stronger temporal encoding. ANOVA showed a significant group effect ($F_{(1,792)} = 36.5$, $p = 2.3 \times 10^{-9}$, $\eta_p^2 = 0.044$) and component effect ($F_{(3,792)} = 6.5$, $p = 2.4 \times 10^{-4}$, $\eta_p^2 = 0.024$). Error bars denote SEM.

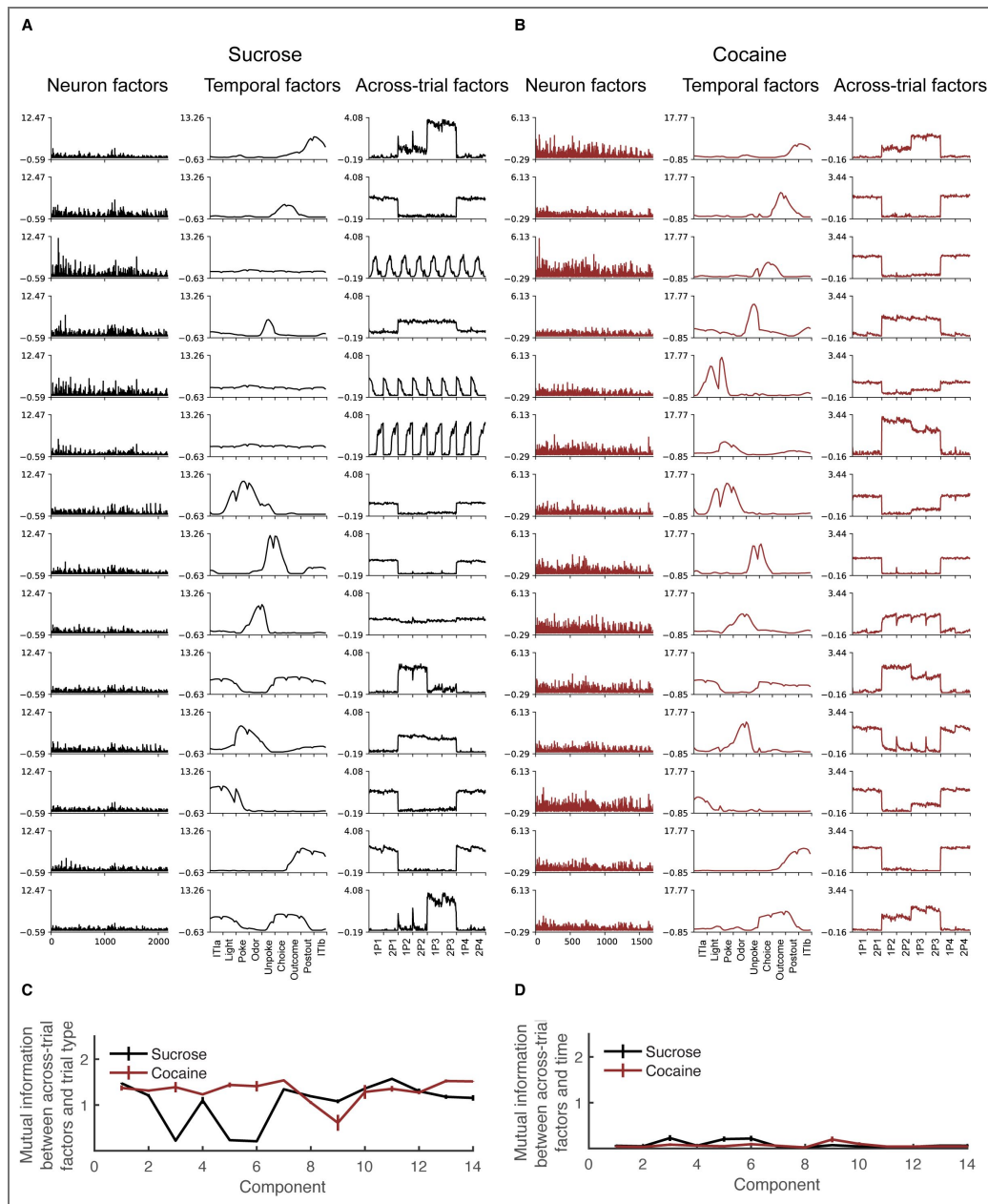


Figure 4—figure supplement 3. Fourteen component model, cocaine use impairs OFC function in temporal cognition and hinders generalization across odor-overlapping, task-irrelevant sequences

(A–B) Fourteen components from TCA applied to Sucrose (A) and Cocaine (B) neural activity are shown. Each component includes a neuron factor (left), a temporal factor spanning nine events with eight time points each (middle), and a trial factor (right). Temporal segments are grouped as: preTrial (1–8), Light (9–16), Poke (17–24), Odor (25–32), Unpoke (33–40), Choice (41–48), Outcome (49–56), postTrial1 (57–64), and postTrial2 (65–72). The dataset comprises eight trial types (40 trials each): rewarded trials (1–80, 241–320; P1, P4) and unrewarded trials (81–240; P2, P3). Low-dimensional components were extracted from a 14-component model for each group. In Sucrose, components 2, 7, 8, 12 and 13 encode positive trials; components 1, 4, 10, 11 and 14 encode negative trials. In Cocaine, positive trial components are 2, 3, 5, 7, 8, 11, 12, and 13; negative trials are encoded by components 1, 4, 6, 9, 10, and 14. Additionally, Sucrose components 3, 5 and 6 encode temporal structure, and component 9 is broadly active across trials. (C) Mutual information (MI) between across-trial factors and trial types is lower in the Sucrose group, particularly at components 3, 5 and 6, indicating weaker encoding of trial identity. A two-way ANOVA confirmed a significant group difference ($F_{(1,308)} = 571.6$, $p = 4.1 \times 10^{-25}$, $\eta_p^2 = 0.29$) and component effect ($F_{(13,308)} = 33.0$, $p = 1.0 \times 10^{-50}$, $\eta_p^2 = 0.58$). (D) MI between across-trial factors and time is higher in Sucrose, especially at components 5 and 9, reflecting stronger temporal encoding. A two-way ANOVA revealed a significant group effect ($F_{(1,2772)} = 3.8$, $p = 0.046$, $\eta_p^2 = 0.0014$) and component effect ($F_{(13,2772)} = 5.5$, $p = 5.6 \times 10^{-10}$, $\eta_p^2 = 0.025$). Error bars represent SEM.

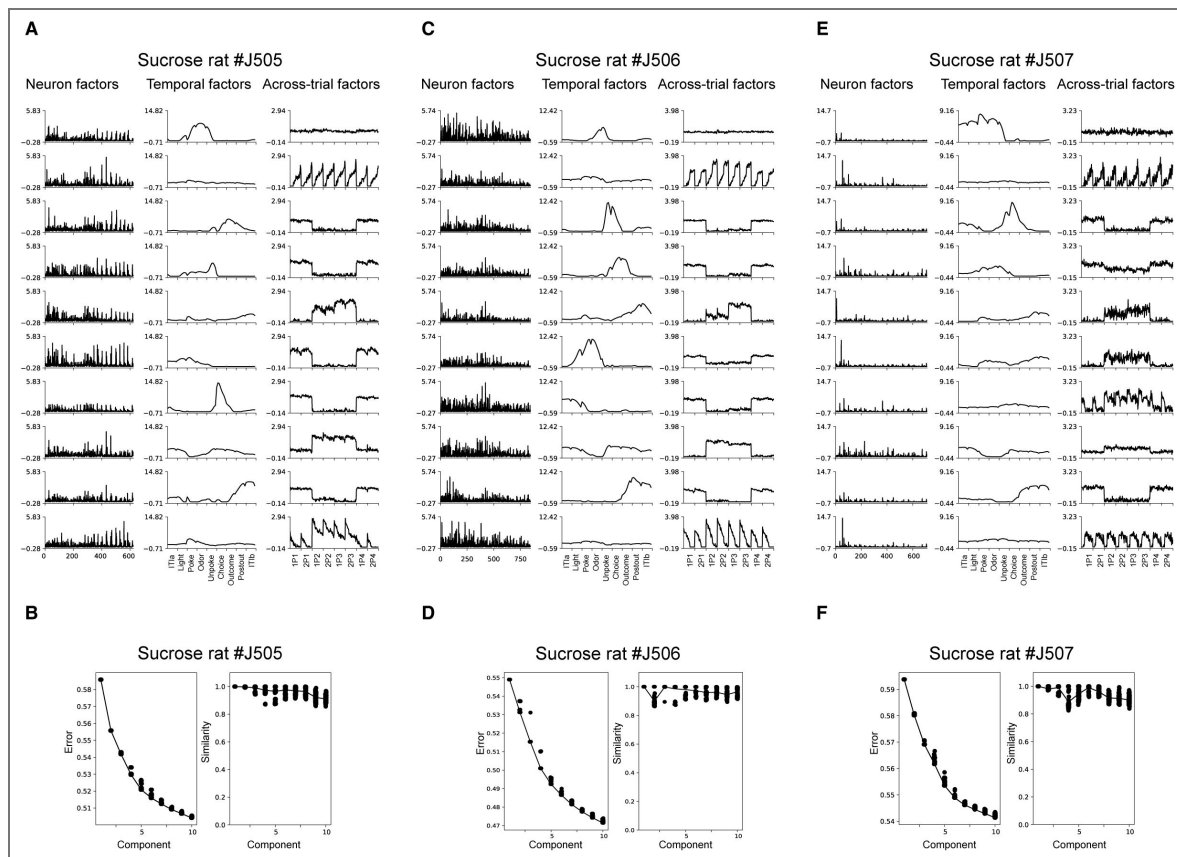


Figure 4—figure supplement 4. Ten-component model for individual Sucrose rats

(A, C, E) Ten-component TCA models are presented for neuronal activity recorded from individual sucrose-trained rats. Each component comprises three factors: a neuron factor (left column), a temporal factor spanning nine task events with eight time points per event (middle column), and a trial factor (right column). Temporal factors are organized as follows: 1–8 (preTrial), 9–16 (Light), 17–24 (Poke), 25–32 (Odor), 33–40 (Unpoke), 41–48 (Choice), 49–56 (Outcome), 57–64 (postTrial1), and 65–72 (postTrial2). The dataset includes eight trial types with 40 trials each. Trial factors 1–80 and 241–320 correspond to rewarded (positive) trials, while 81–240 correspond to unrewarded (negative) trials. Trials are further categorized by position: 1–80 (P1), 81–160 (P2), 161–240 (P3), and 241–320 (P4). These low-dimensional components reflect the latent structure of population activity. Across Sucrose rats #J505, #J506, and #J507, several TCA components consistently distinguished trial valence. Components 3, 4, and 9 were associated with positive trials in all three rats, with components 6 and 7 additionally contributing in #J505 and #J506. Negative trials were reliably encoded by components 5 and 8 across all rats, with component 6 also contributing in #J507. Furthermore, each rat exhibited temporal components (typically 2 and 10, with 7 also involved in #J507) and a broadly active early component (1) that was consistent across trial types. (B, D, F) Error plots (left panels) and similarity plots (right panels) are shown for each Sucrose rat. The error plots demonstrate that all 100 runs with a fixed number of components yielded nearly identical reconstruction errors, indicating stable model convergence. Models with 9 and 10 components produced similarly low errors, with minimal improvement beyond 10 components. The similarity plots quantify consistency across model runs. For each component number, similarity was computed relative to the best-fit model, and mean similarity is plotted as a function of component number. Across all tested component counts, repeated runs produced high similarity scores (generally > 0.8), indicating that the decomposition was highly stable and reproducible for all Sucrose rats.

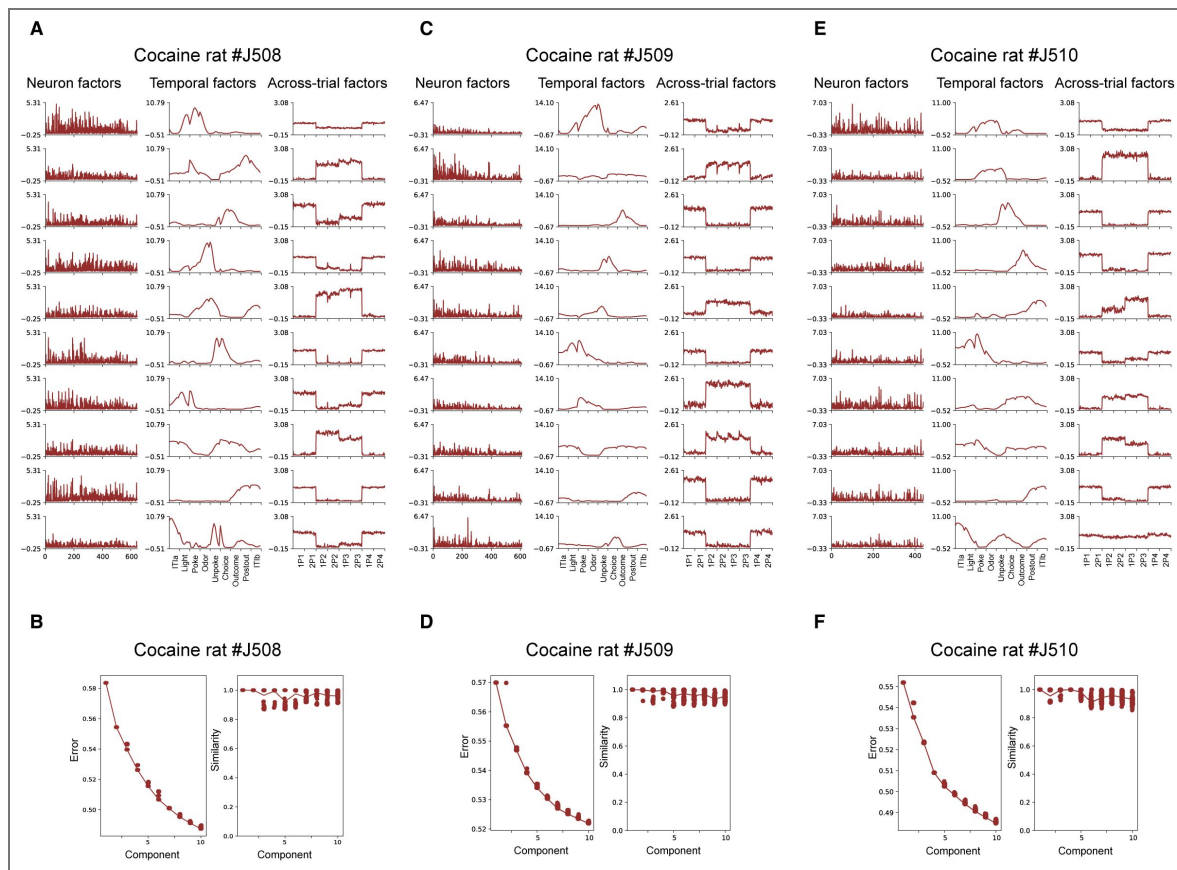


Figure 4—figure supplement 5. Ten-component TCA model applied to individual rats from the Cocaine group

(A, C, E) TCA-derived ten-component models are shown for individual Cocaine rats. Each component includes a neuron factor (left), a temporal factor spanning nine task events (middle), and a trial factor (right). Temporal factors cover preTrial (1–8), Light (9–16), Poke (17–24), Odor (25–32), Unpoke (33–40), Choice (41–48), Outcome (49–56), postTrial1 (57–64), and postTrial2 (65–72). The dataset includes eight trial types (40 trials each): trials 1–80 and 241–320 are rewarded (positive), and 81–240 are unrewarded (negative), grouped by position (P1–P4). In the Cocaine rats, positive trials were consistently encoded by components 1, 3, 4, 6, 9, and 10. Negative trials were represented by components 2, 5, and 8 across all Cocaine rats, with component 7 additionally contributing to rats #J509 and #J510. (B, D, F) Error plots (left) and similarity plots (right) are shown for each Cocaine rat. The error plots indicate that across 100 runs with a fixed number of components, reconstruction errors were nearly identical, demonstrating stable model convergence. Models with 9 or 10 components achieved comparably low errors, with little improvement beyond 10 components. The similarity plots quantify consistency across runs: for each component count, similarity was calculated relative to the best-fit model, and mean similarity is plotted as a function of component number. Across all component counts, repeated runs yielded high similarity scores (generally >0.8), confirming that the decomposition was stable and reproducible in all sucrose rats.

Data availability

Data and code availability: All data and analysis code associated with this study are available on OSF at <https://osf.io/azvhm/overview>.

Acknowledgements

This research was funded by the Intramural Research Program at the National Institute on Drug Abuse (ZIA-DA000587). The views expressed in this article are solely those of the authors and do not necessarily represent the opinions of the NIH or DHHS.

Additional information

Author contributions

J.Z. and G.S. designed the experiments, while J.Z. and W.Z. collected and analyzed the data with guidance and technical support from Z.Z. and L.E.M.. W.Z., J.Z., and G.S. wrote the manuscript, incorporating feedback from all authors.

Funding

Funder	Grant reference number	Author
HHS NIH National Institute on Drug Abuse (NIDA)	Z1A-DA000587	Geoffrey Schoenbaum

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this study, the authors trained rats on a "figure 8" go/no-go odor discrimination task. Six odor cues (3 rewarded and 3 non-rewarded) were presented in a fixed temporal order and arranged into two alternating sequences that partially overlap (Sequence #1: 5⁺-0⁻-1⁻-2⁺; Sequence #2: 3⁺-0⁻-1⁻-4⁺) –forming an abstract figure-8 structure of looping odor cues.

This task is particularly well-suited for probing representations of hidden states, defined here as the animal's position within the task structure beyond superficial sensory features. Although the task can be solved without explicit sequence tracking, it affords the opportunity to generalize across functionally equivalent trials (or "positions") in different sequences, allowing the authors to examine how OFC representations collapse across latent task structure.

Rats were first trained to criterion on the task and then underwent 15 days of self-administration of either intravenous cocaine (3 h/day) or sucrose. Following self-administration, electrodes were implanted in lateral OFC, and single-unit activity was recorded while rats performed the figure-8 task.

Across a series of complementary analyses, the authors report several notable findings. In control animals, IOFC neurons exhibit representational compression across corresponding positions in the two sequences. This compression is observed not only in trial/positions involving overlapping odor (e.g., Position 3 = odor 1 in sequence 1 vs sequence 2), but also in trials/positions involving distinct, sequence-specific odors (e.g., Position 4: odor 2 vs odor 4) -- indicating generalization across functionally equivalent task states. Ensemble decoding confirms that sequence identity is weakly decodable at these positions, consistent with the idea that OFC representations collapse incidental differences in sensory information into a common latent or hidden state representation. In contrast, cocaine-experienced rats show persistently stronger differentiation between sequences, including at overlapping odor positions.

Strengths:

- Elegant behavioral design that affords the detection of hidden-state representations.
- Sophisticated and complementary analytical approaches (single-unit activity, population decoding, and tensor component analysis).

Weaknesses:

- The number of subjects is small --can't fully rule out idiosyncratic, animal-specific effects.

Comments on revisions:

The authors have thoroughly addressed all of my previous comments. Congratulations on an excellent paper!

<https://doi.org/10.7554/eLife.109883.2.sa2>

Reviewer #2 (Public review):

In the current study, the authors use an odor-guided sequence learning task described as a "figure 8" task to probe neuronal differences in latent state encoding within the orbitofrontal cortex after cocaine ($n = 3$) vs sucrose ($n = 3$) self-administration. The task uses six unique odors which are divided into two sequences that run in series. For both sequences, the 2nd and 3rd odors are the same and predict reward is not available at the reward port. The 1st and 4th odors are unique, and are followed by reward. Animals are well-trained before undergoing electrode implant and catheterization, and then retrained for two weeks prior to recording. The hypothesis under test is that cocaine-experienced animals will be less able to use the latent task structure to perform the task, and instead encode information about each unique sequence that is largely irrelevant. Behaviorally, both cocaine and sucrose-experienced rats show high levels of accuracy on task, with some group differences noted. When comparing reaction times and poke latencies between sequences, more variability was observed in the cocaine-treated group, implying animals treated these sequences somewhat differently. Analyses done at the single unit and ensemble level suggests that cocaine self-administration had increased the encoding of sequence-specific information, but decreased generalization across sequences. For example, the ability to decode odor position and sequence from neuronal firing in cocaine-treated animals was greater than controls. This pattern resembles that observed within the OFC of animals that had fewer training sessions. The authors then conducted tensor component analysis (TCA) to enable a more "hypothesis agnostic" evaluation of their data.

Overall, the paper is well written and the authors do a good job of explaining quite complicated analyses so that the reader can follow their reasoning. The findings are important, and the results are compelling. The introduction and discussion contextualize the experiments in the context of the literature, and explain the novelty and significance of the current findings. Specifically, the observation that cocaine self-administration impairs generalization across task sequences at the single unit level builds on previous observations of aberrant neuronal activity within the OFC in animals with a history of cocaine self-administration. These new data point to a neurophysiological mechanism that could explain why drug-seeking is so context dependent, and hard to ameliorate with therapeutic strategies that take place within a clinical setting.

The authors clearly acknowledge the major limitations of this work, namely that the sample size is restricted due to the technical challenges of performing in vivo electrophysiology recordings combined with self-administration, and that animals of only one sex were used. Importantly, the data from all rats within each group was remarkably homogeneous, increasing confidence in the conclusions drawn.

<https://doi.org/10.7554/eLife.109883.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

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Elegant behavioral design that affords the detection of hidden-state representations.

Sophisticated and complementary analytical approaches (single-unit activity, population decoding, and tensor component analysis).

Weaknesses:

The number of subjects is small - can't fully rule out idiosyncratic, animal-specific effects.

Comments

(1) Emergence of sequence-dependent OFC representations across learning.

A conceptual point that would benefit from further discussion concerns the emergence of sequence-dependent OFC activity at overlapping positions (e.g., position P3, odor 1). This implies knowledge of the broader task structure. Such representations are presumably absent early in learning, before rats have learned the sequence structure. While recordings were conducted only after rats were well trained, it would be informative if the authors could comment on how they envision these representations developing over learning. For example, does sequence differentiation initially emerge as animals learn the overall task structure, followed by progressive compression once animals learn that certain states are functionally equivalent? Clarifying this learning-stage interpretation would strengthen the theoretical framing of the results.

We agree that the emergence of sequence-dependent OFC activity at overlapping positions (e.g., P3) implies knowledge of the broader task structure and therefore must depend on learning. Although we did not record during early acquisition in the current study, we can outline a learning-stage framework consistent with both prior work and the comparative analyses included here and include it in the discussion.

We think the development of OFC representations is a multi-stage process. Early in learning, before animals have acquired the sequential structure of the task, OFC activity is likely

dominated by local sensory features and immediate reinforcement history, with little differentiation between sequences at overlapping positions. As animals learn that odors are embedded within extended sequences that have utility for predicting future outcomes, OFC representations would begin to differentiate identical sensory cues based on their sequence context, giving rise to sequence-dependent activity at positions such as P3. This stage reflects acquisition of the broader task structure and the recognition that current cues carry information about future states.

With continued training, however, OFC representations normally undergo a further refinement: positions that differ in sensory identity but are functionally equivalent become compressed, while distinctions that are irrelevant for guiding behavior are suppressed. Evidence for this later stage comes from our over-trained control animals, in which discrimination between overlapping positions is near chance across most trial epochs, and from prior work using the same task in less-trained animals, where sequence-dependent discrimination is more strongly preserved. Thus, sequence differentiation appears to emerge during structure learning but is subsequently down weighted as animals learn which distinctions are behaviorally irrelevant.

Within this framework, prior cocaine exposure appears to interfere specifically with this later refinement stage. Cocaine-experienced rats exhibit OFC representations resembling those seen earlier in learning—retaining sequence-dependent discrimination at overlapping and functionally equivalent positions—despite extensive training. This suggests not a failure to acquire task structure per se, but rather an impairment in the ability to collapse across states that share common underlying causes.

(2) Reference to the 24-odor position task

The reference to the previously published 24-odor position task is not well integrated into the current manuscript. Given that this task has already been published and is not central to the main analyses presented here, the authors may wish to a) better motivate its relevance to the current study or b) consider removing this supplemental figure entirely to maintain focus.

Thanks for your suggestion, we have removed this supplemental figure as suggested.

(3) Missing behavioral comparison

Line 117: the authors state that absolute differences between sequences differ between cocaine and sucrose groups across all three behavioral measures. However, Figure 1 includes only two corresponding comparisons (Fig. 1I-J). Please add the third measure (% correct) to Figure 1, and arrange these panels in an order consistent with Figure 1F-H (% correct, reaction time, poke latency).

Thanks for your suggestion, we have included the related figure as suggested.

(4) Description of the TCA component

Line 220: authors wrote that the first TCA component exhibits low amplitude at positions P1 and P4 and high amplitude at positions P2 and P3. However, Figure 3 appears to show the opposite pattern (higher magnitude at P1 and P4 and lower magnitude at P2 and P3). Please check and clarify this apparent discrepancy. Alternatively, a clearer explanation of how to interpret the temporal dynamics and scaling of this component in the figure would help readers correctly understand the result.

Thanks for your suggestion. We appreciate this point and agree that clearer guidance on how to interpret the temporal and scaling properties of the tensor components would help readers. In the TCA framework, each component is defined by three separable factors: a

neuron factor, a temporal factor, and a trial (position) factor. The temporal factor reflects the shape of the activity pattern within a trial, indicating when during the trial that component is expressed, whereas the trial factor reflects how strongly that temporal pattern is expressed at each position and across trials.

Importantly, the absolute scaling of these factors is not independently meaningful. Because TCA components are scale-indeterminate, the magnitude of the temporal factor and the trial factor should be interpreted relative to one another within a component, not across components. Thus, a large value in the trial factor does not imply stronger neural activity per se, but rather greater expression of that component's characteristic temporal pattern at that position or trial.

Accordingly, when a component shows similar temporal dynamics across groups but differs in its trial factor structure—as observed here—the interpretation is that the same within-trial dynamics are being differentially recruited across task positions, rather than that the timing of neural responses has changed.

We have added a brief discussion of this in this section of the results in the manuscript.

(5) Sucrose control

Sucrose self-administration is a reasonable control for instrumental experience and reward exposure, but it means that this group also acquired an additional task involving the same reinforcer. This experience may itself influence OFC representations and could contribute to the generalization observed in control animals. A brief discussion of this possibility would help contextualize the interpretation of cocaine-related effects.

We agree that sucrose self-administration is not a perfect neutral manipulation and that this experience could, in principle, influence OFC representations. In particular, sucrose self-administration involves instrumental responding for the same primary reinforcer used in the odor task, and thus may promote additional learning about reward predictability, action–outcome contingencies, or contextual structure that could facilitate generalization.

Several considerations, however, suggest that the generalization observed in control animals primarily reflects learning-dependent refinement of task representations rather than a specific consequence of sucrose self-administration per se. First, the amount of sucrose administered during this phase was minimal (50 μ l $\times$ 60 presses at most per session for 14 sessions) compared with the total sucrose reward obtained during task recording (100 μ l $\times$ 160 trials per session for several dozen sessions). Second, all rats were extensively trained on the odor sequence task prior to any self-administration, and the key signatures of compression and generalization we report—near-chance discrimination between functionally equivalent positions—are consistent with prior studies using the same task in animals that did not undergo sucrose self-administration. Finally, comparisons to less-trained animals in earlier work show that OFC representations evolve toward greater abstraction with increasing task experience, indicating that generalization is a property of advanced learning rather than a unique outcome of sucrose exposure.

Importantly, even if sucrose self-administration were to enhance generalization in OFC, this would not account for the primary finding that cocaine-experienced rats fail to show these signatures despite identical task training and parallel instrumental experience. Thus, the critical comparison is not between sucrose-trained animals and naive controls, but between two groups matched for self-administration experience, differing only in the pharmacological consequences of the reinforcer. Within this framework, the absence of position-general representations in cocaine-experienced rats reflects a disruption of normal learning-dependent abstraction rather than an artifact of the control condition.

We have added a brief discussion acknowledging that sucrose self-administration may bias OFC toward abstraction, while emphasizing that cocaine exposure prevents the emergence or maintenance of these representations under otherwise comparable experiential conditions.

(6) Acknowledge low N

The number of rats per group is relatively low. Although the effects appear consistent across animals within each group, this sample size does not fully rule out idiosyncratic, animal-specific effects. This limitation should be explicitly acknowledged in the manuscript.

We acknowledge that the number of animals per group is relatively small and therefore cannot fully rule out animal-specific effects. However, the key neural and behavioral signatures reported here were consistent across individual animals within each group and across multiple levels of analysis, and no outliers were observed. In addition, sample sizes of this scale are common in cocaine self-administration studies due to their technical and logistical constraints. We did not attempt to obscure this limitation and have now explicitly acknowledged it in the manuscript discussion.

(7) Figure 3E-F: The task positions here are ordered differently (P1, P4, P2, P3) than elsewhere in the paper. Please reorder them to match the rest of the paper.

Thank you for pointing this out. We agree that the ordering of task positions in Figures 3E-F should be consistent with the rest of the manuscript. We have reordered the positions to match the standard sequence order used elsewhere in the paper (P1, P2, P3, P4) to improve clarity and avoid confusion.

Reviewer #2 (Public review):

In the current study, the authors use an odor-guided sequence learning task described as a "figure 8" task to probe neuronal differences in latent state encoding within the orbitofrontal cortex after cocaine (n = 3) vs sucrose (n = 3) self-administration. The task uses six unique odors which are divided into two sequences that run in series. For both sequences, the 2nd and 3rd odors are the same and predict reward is not available at the reward port. The 1st and 4th odors are unique, and are followed by reward. Animals are well-trained before undergoing electrode implant and catheterization, and then retrained for two weeks prior to recording. The hypothesis under test is that cocaine-experienced animals will be less able to use the latent task structure to perform the task, and instead encode information about each unique sequence that is largely irrelevant. Behaviorally, both cocaine and sucrose-experienced rats show high levels of accuracy on task, with some group differences noted. When comparing reaction times and poke latencies between sequences, more variability was observed in the cocaine-treated group, implying animals treated these sequences somewhat differently. Analyses done at the single unit and ensemble level suggests that cocaine self-administration had increased the encoding of sequence-specific information, but decreased generalization across sequences. For example, the ability to decode odor position and sequence from neuronal firing in cocaine-treated animals was greater than controls. This pattern resembles that observed within the OFC of animals that had fewer training sessions. The authors then conducted tensor component analysis (TCA) to enable a more "hypothesis agnostic" evaluation of their data.

Overall, the paper is well written and the authors do a good job of explaining quite complicated analyses so that the reader can follow their reasoning. I have the following comments.

While well-written, the introduction mainly summarises the experimental design and results, rather than providing a summary of relevant literature that informed the experimental design. More details regarding the published effects of cocaine self-administration on OFC firing, and on tests of behavioral flexibility across species, would ground the paper more thoroughly in the literature and explain the need for the current experiment.

We appreciate this suggestion and have tried to expand the Introduction to more explicitly situate the study within the existing literature on cocaine-induced changes in OFC function. In particular, prior work has shown that cocaine self-administration alters OFC firing properties and disrupts behavioral flexibility across species, including impairments in reversal learning, outcome devaluation, and sensory preconditioning. We have revised the Introduction to expand this literature review and more clearly articulate how these established findings motivated our focus on OFC representations of hidden task structure and generalization.

For Fig 1F, it is hard to see the magnitude of the group difference with the graph showing 0-100%- can the y axis be adjusted to make this difference more obvious? It looks like the cocaine-treated animals were more accurate at P3- is that right?

The concluding section is quite brief. The authors suggest that the failure to generalize across sequences observed in the current study could explain why people who are addicted to cocaine do not use information learned e.g. in classrooms or treatment programs to curtail their drug use. They do not acknowledge the limitations of their study e.g. use of male rats exclusively, or discuss alternative explanations of their data.

We agree that the current 0–100% scale can make small differences difficult to discern. We will make it clear in the figure captions (We will adjust the y-axis to a narrower range to better highlight group differences). Across P3, cocaine-experienced rats were more accurate than controls.

We appreciate the suggestion to expand the discussion. We have revised the concluding section to acknowledge key limitations, including the use of only male rats, the number of subjects, and to note that alternative explanations—such as differences in motivational state or attention—could also contribute to the observed effects. These revisions provide a more balanced interpretation while retaining the focus on OFC-mediated generalization as a potential mechanism for persistent, context-specific drug-seeking.

Is it a problem that neuronal encoding of the "positions" i.e. the specific odors was at or near chance throughout in controls? Could they be using a simpler strategy based on the fact that two successive trials are rewarded, then two successive trials are not rewarded, such that the odors are irrelevant?

We thank the reviewer for this point. While neuronal encoding of individual positions (specific odors) in control animals was comparatively lower, this does not indicate that the rats were using a simpler strategy based solely on reward patterns. First, rats were extensively trained on the odor sequence task prior to recordings, demonstrating accurate discrimination across all positions, and their trial-by-trial behavior reflects sensitivity to specific odors rather than only reward alternation. Second, the task design—with overlapping sequences and positions that differ in reward contingency across sequences—requires tracking odor-specific context to maximize reward; a purely “two rewarded, two non-rewarded” strategy would fail at overlapping positions and would not account for the compression of functionally equivalent positions observed in the OFC. Third, in the less-trained rats shown in Figure 3C, decoding accuracy was higher than in the sucrose group, indicating that these animals still differentiated negative positions. With additional training, decoding patterns suggested improved generalization across positions. Thus, the near-chance

neural selectivity in controls reflects representation of latent task states rather than external sensory cues, consistent with the idea that OFC abstracts task-relevant structure and ignores irrelevant sensory differences.

When looking at the RT and poke latency graphs, it seems the cocaine-experienced rats were faster to respond to rewarded odors, and also faster to poke after P3. Does this mean they were more motivated by the reward?

At present, the basis of these response-time differences remains unclear, in part because motivation is difficult to define operationally. If motivation is indexed solely by reaction time or poke latency, then the data are consistent with increased response vigor in cocaine-experienced rats. Indeed, RT and poke-latency measures indicate that cocaine-experienced rats responded more quickly on some rewarded trials, including after P3. However, overall task performance was high in both groups, suggesting that these differences cannot be attributed simply to superior learning or engagement. Faster responses may also reflect differences in deliberation or strategy, with cocaine-experienced rats relying more on rapid, stimulus-driven responding and sucrose-trained rats engaging in more careful evaluation. In addition, altered reward sensitivity or persistent effects of cocaine exposure may contribute to these behavioral differences. Thus, the faster responses observed in cocaine-experienced rats likely reflect a combination of heightened reward responsivity and altered encoding of task structure, rather than a straightforward increase in motivation alone.

Recommendations for the authors:

The reviewers were very positive about the manuscript and emphasized the rigor and state of the art analyses. Two points that came up were the very small n (6 total and 3 per condition) and the exclusive use of males. Adding more subjects is not recommended. However, more discussion and acknowledgement of this issue is recommended. The main concern is that idiosyncratic differences between individuals (not differences in cocaine history) are responsible for the differences observed in OFC encoding.

We acknowledge that the sample size ($n = 3$ per group) and use of only male rats limit generalizability and do not fully rule out idiosyncratic, individual-specific effects. However, the key neural and behavioral signatures we report were consistent across all animals within each group and across multiple analyses (single-unit, ensemble decoding, and TCA). We now explicitly note these limitations in the Discussion, emphasizing that while individual variability cannot be fully excluded, the convergence of results across multiple levels of analysis supports the interpretation that the observed differences reflect effects of prior cocaine exposure rather than idiosyncratic differences.

Reviewer #2 (Recommendations for the authors):

In the legend to figure 2, the authors state "Notably, rats could discriminate between the two sequences (S1 vs. S2) based solely on current sensory information at two task epochs ["Odor" at P3 and P4; black bars]. At all other task epochs, indicated by gray bars, the discrimination relied on an internal memory of events". I'm confused by this statement—how does the odor at P3 help to discriminate the sequences? Surely P1 and P4 are the times when the odor sampling indicates which sequence they are in?

We thank the reviewer for pointing out this source of confusion. The statement in the original figure legend was imprecise, and we have removed the figure and revised the figure legends because the results in the left panel substantially overlapped with those shown in the right panel. In this task, odors at positions P1 and P4 are the only cues that directly signal sequence identity, whereas the odors presented at P2 and P3 are identical across sequences. Accordingly, discrimination observed during the “Odor” epoch at P3 does not reflect sensory differences but instead depends on the animal’s use of internal memory or sequence context to infer sequence identity.

<https://doi.org/10.7554/eLife.109883.2.sa0>